
Classifying Representations of $\mathfrak{sl}_3\mathbb{C}$

A Model Example for the Representation Theory of Semisimple Lie Algebras

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Primer on Lie Algebras and Representation Theory

Representations of $\mathfrak{sl}_2\mathbb{C}$

Classifying Representations of $\mathfrak{sl}_3\mathbb{C}$

Some Examples

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- The Jacobi identity:

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$$

for all $X, Y, Z \in \mathfrak{g}$ and all $\alpha, \beta \in F$.

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- ▶ The set of anti-symmetric matrices with entries in $\mathbb{C}$ and trace zero forms a Lie algebra with the commutator as its Lie bracket, denoted $\mathfrak{so}_n\mathbb{C}$

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Representations of simply connected Lie groups are in one-to-one correspondence with representations of their Lie algebras.

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- ▶ A subspace W of V is *invariant* if $\rho(X)w \in W$ for all $w \in W$ and $X \in \mathfrak{g}$
- ▶ A nonzero representation is *irreducible* if the only invariant subspaces are V and $\{0\}$

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- The *standard representation* $\mathfrak{g} \hookrightarrow \mathfrak{gl}(V)$

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- ▶ We choose¹ the following basis for $\mathfrak{sl}_2\mathbb{C}$:

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

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- Note, $[H, X] = 2X$, $[H, Y] = -2Y$, $[X, Y] = H$

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- ▶ If $v \in V_{\alpha}$, then $X(v) \in V_{\alpha+2}$ and $Y(v) \in V_{\alpha-2}$
- ▶ Since V is finite dimensional and each $\alpha \in \mathbb{R}$ for all α ,² there is a *highest weight* n such that $X(v) = 0$ for all $v \in V_n$

² Why?

- For some highest weight vector $v \in V_n$, the vectors

$$\{v, Y(v), Y^2(v), \dots\}$$

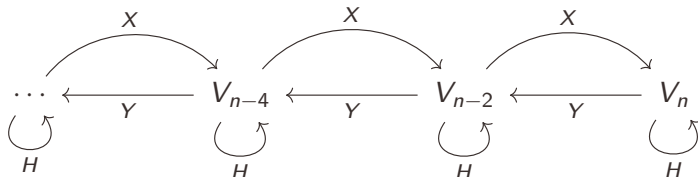
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 - Each weight space V_α is one dimensional
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- In fact, irreducible representations of $\mathfrak{sl}_2\mathbb{C}$ are *entirely classified* by this choice of highest weight n
- ▶ It turns out that any irreducible representation of $\mathfrak{sl}_2\mathbb{C}$ is a symmetric power $\text{Sym}^n V$ of the standard representation $V \cong \mathbb{C}^2$

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 - Namely, $\mathfrak{h}$ is the two-dimensional subalgebra of all diagonal matrices³

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 - Namely, $\mathfrak{h}$ is the two-dimensional subalgebra of all diagonal matrices³
- ▶ This gives us a natural way to talk about the decomposition of $V = \bigoplus V_\alpha$

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- ▶ An *eigenspace* associated to α is the subspace of all eigenvectors $v \in V$ with eigenvalue α

Any finite-dimensional representation V of $\mathfrak{sl}_3\mathbb{C}$ decomposes as

$$V = \bigoplus_{\alpha} V_{\alpha}$$

where V_{α} is an eigenspace for $\mathfrak{h}$ and α ranges over a finite subset of $\mathfrak{h}^*$.

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- ▶ This yields a decomposition of $\mathfrak{sl}_3\mathbb{C}$ as a representation of itself:

$$\mathfrak{sl}_3\mathbb{C} = \mathfrak{h} \oplus \left(\bigoplus \mathfrak{g}_\alpha \right)$$

where α ranges over a finite subset of $\mathfrak{h}^*$ and

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- The weights α of the adjoint representation are called the *roots*, the $\mathfrak{g}_\alpha$ are the *root spaces*, and weight vectors are *root vectors*

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$$\mathfrak{h} = \left\{ \left(\begin{pmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix} \mid a_1 + a_2 + a_3 = 0 \right) \right\},$$

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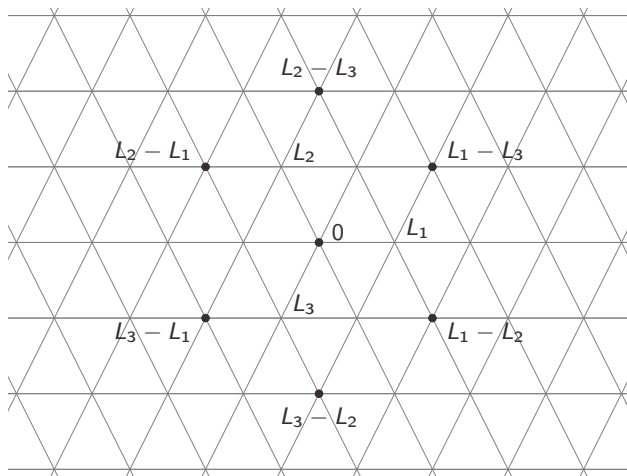
Now, if we take $E_{i,j}$ to be the 3×3 matrix whose (i,j) th entry is 1 with all other entries 0...

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There are thus six functionals $L_i - L_j \in \mathfrak{h}^*$ appearing in the noted decomposition, with $\mathfrak{g}_{L_i - L_j}$ generated by $E_{i,j}$:

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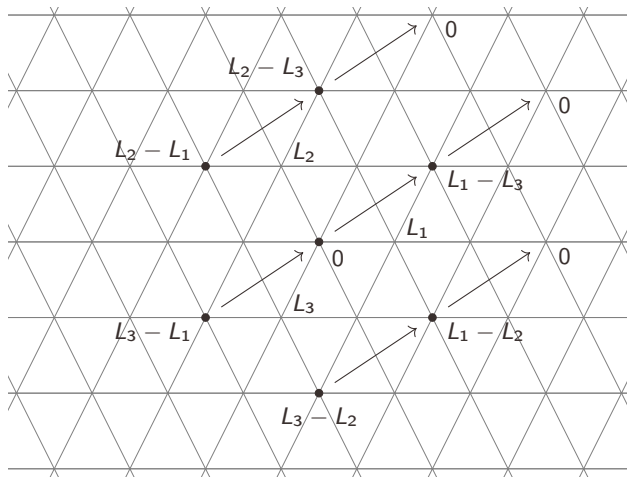
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- Thus, $[X, Y] = \text{ad}(X)(Y)$ is an eigenvector for $\mathfrak{h}$ with eigenvalue $\alpha + \beta$

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For example, the action of $g_{L_1-L_3}$ may be pictured as



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- Thus, $X(v)$ is again an eigenvector for the action of $\mathfrak{h}$ with eigenvalue $\alpha + \beta$
- So, the action of $\mathfrak{g}_\alpha$ carries V_β to $V_{\beta+\alpha}$

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- ▶ All α lie in some translation of this lattice

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- ▶ But it's more finicky to pick one

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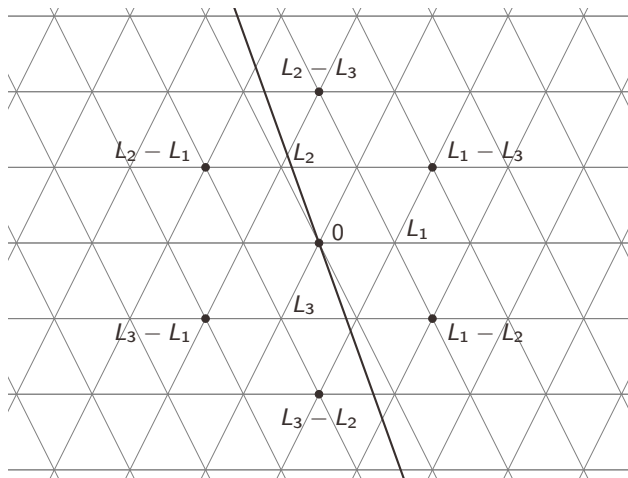
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- ▶ This gives us an eigenvector $v \in V_\alpha$ which is killed simultaneously by $\mathfrak{g}_\beta$ for every β such that $l(\beta) > 0$

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- ▶ This gives us an eigenvector $v \in V_\alpha$ which is killed simultaneously by $\mathfrak{g}_\beta$ for every β such that $l(\beta) > 0$
 - i.e. v is killed by half of the root spaces $\mathfrak{g}_\beta$

"Choosing" a Highest Weight



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For $i < j$, the matrices $E_{i,j}$ generate the positive root spaces and the $E_{j,i}$ generate the negative root spaces. Let

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and let V be an irreducible, finite-dimensional representation of $\mathfrak{sl}_3\mathbb{C}$.

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For any representation V of $\mathfrak{sl}_3\mathbb{C}$, we call this a *highest weight vector*.

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Corollary

The eigenvalues of $H_{1,2}$ on some representation W of $\mathfrak{sl}_{L_1-L_2} \cong \mathfrak{sl}_2\mathbb{C}$ are integral and symmetric with respect to zero.

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Theorem

For any pair of natural numbers a, b , there exists a unique irreducible, finite-dimensional representation $\Gamma_{a,b}$ of $\mathfrak{sl}_3\mathbb{C}$ with highest weight $aL_1 - bL_3$.

Primer on Lie Algebras and Representation Theory

Representations of $\mathfrak{sl}_2\mathbb{C}$

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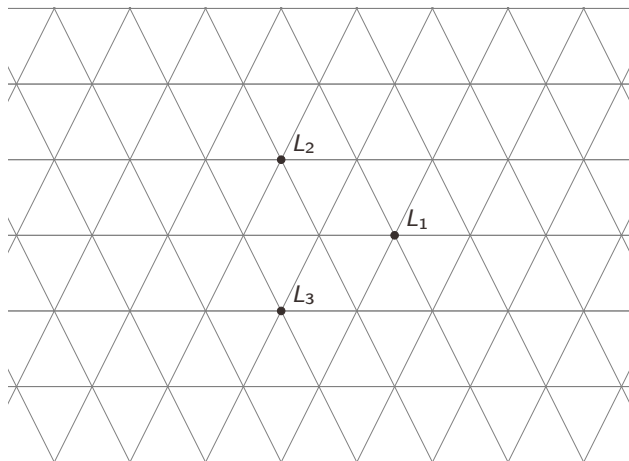
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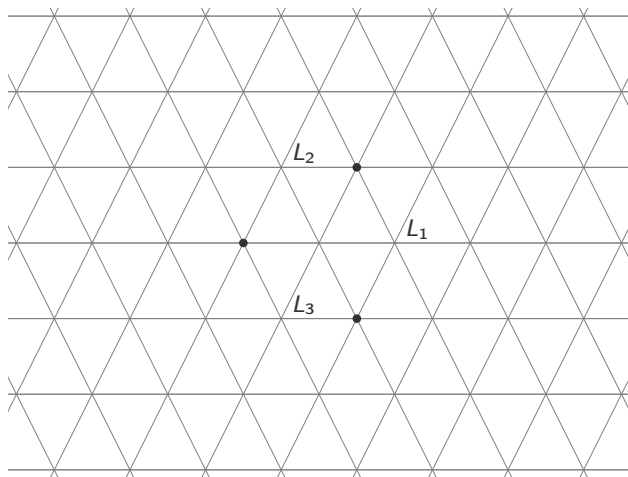
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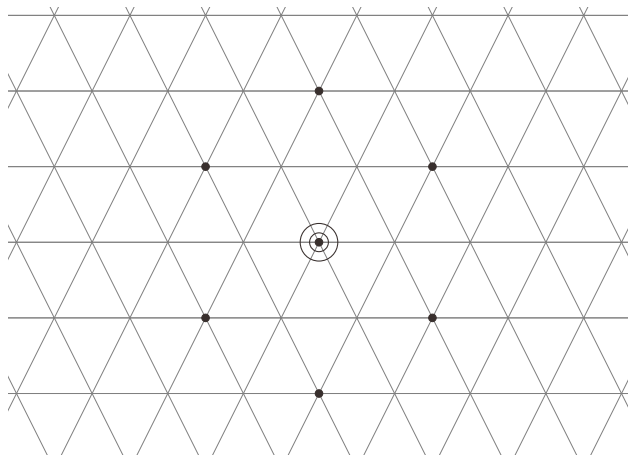
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Thanks for listening!

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- ▶ Thanks to all the DRP organizers!
- ▶ Special thanks to my mentor, Alberto!

Reference: Fulton & Harris, *Representation Theory*